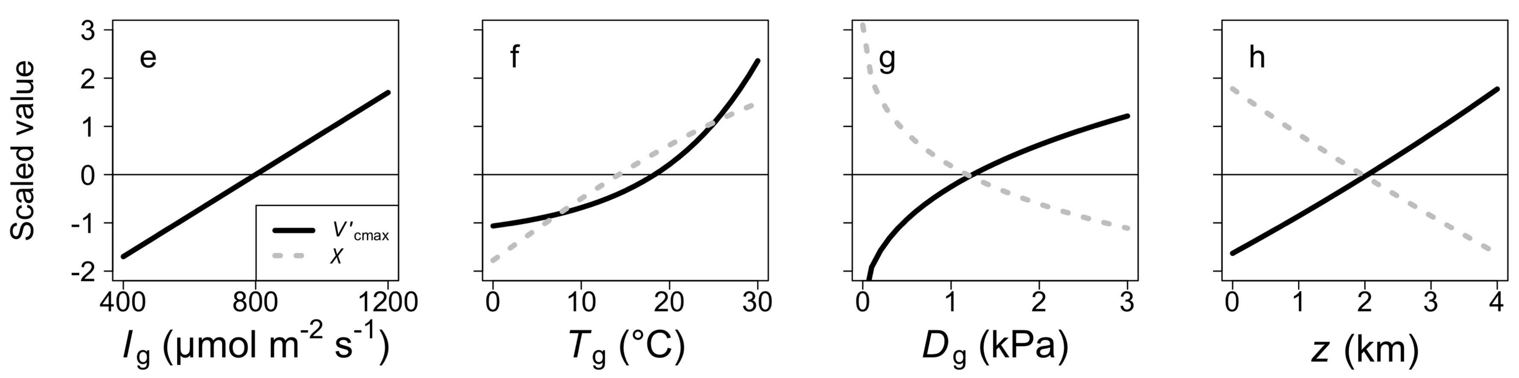
# Global patterns of optimal photosynthetic strategies


Nick Smith (nick.smith@ttu.edu)
Texas Tech University


## (1) Eco-evolutionary optimality (EEO) theory can predict photosynthetic traits

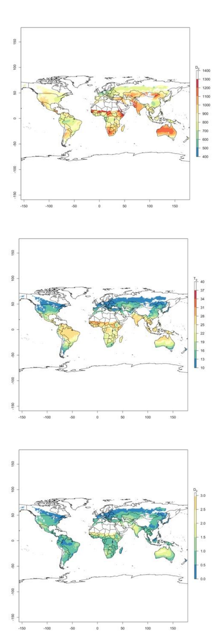
EEO core assumptions:

1. Plants will strive to maintain the fastest photosynthetic rates at the lowest costs
2. Photosynthetic costs are related to carbon, nitrogen, and water acquisition and use
3. Climate and resource availability influence costs and benefits




Example of EEO predictions for trait-environment relationships as presented in Smith et al. (2019). Key: $V_{cmax}$=maximum rate of RuBisCO carboxylation at growth temperature, χ=intercellular to atmospheric $CO_2$ ratio, $I_g$=growing season PAR, $T_g$=growing season temperature, $D_g$=growing season VPD, z=elevation.

## (2) I simulated multi-trait *photosynthetic strategies* at global and local scales

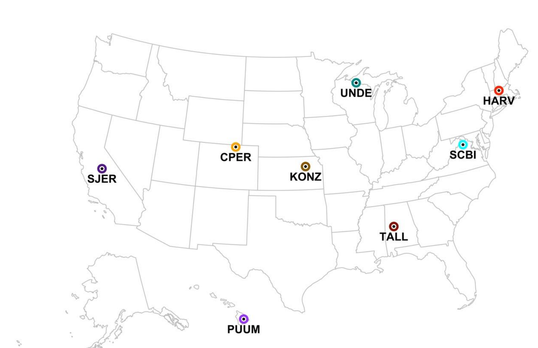


Global-scale simulations

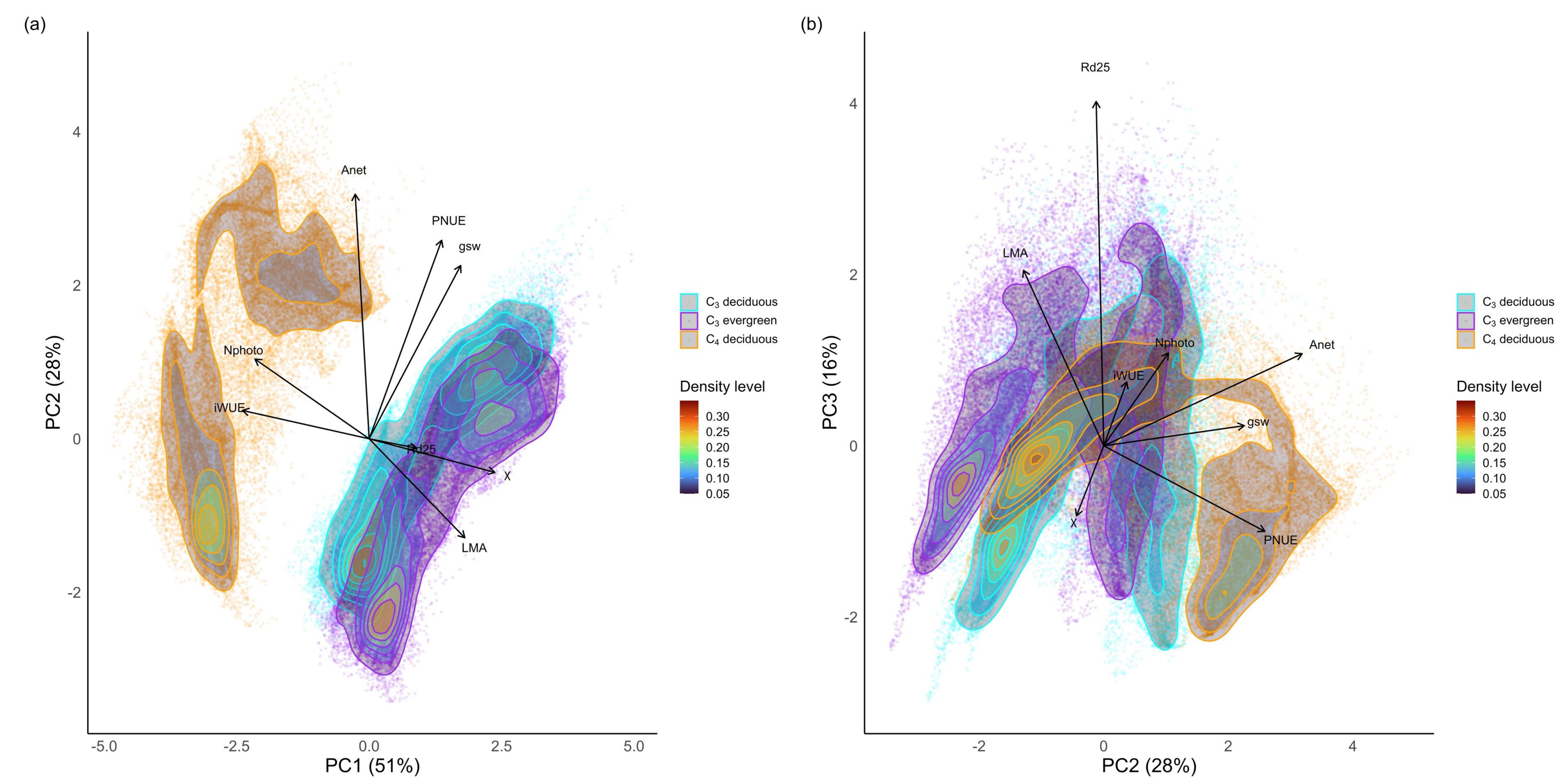
- Simulated optimal photosynthetic traits at 0.5° land surface gridcells
- Climate data was 1901-2015 growing season averages from CRU TS3.24.01 and WFDEI
- Constant resource use costs
- Analyses considered traits related to carbon uptake ($A_{net}$), leaf construction and maintenance costs ($R_{d25}$, LMA), photosynthetic resource use ($N_{photo}$, $g_{sw}$), and photosynthetic efficiency (iWUE, PNUE)

Local-scale simulations

- Simulations at 8 NEON sites, representing four biomes ($C_4$ deciduous, $C_3$ deciduous, $C_3$ evergreen, $C_3$ mixed)
- 1000 simulations per site with climate and soil resource costs sampled from normal distribution
  - Climate: mean ± 0.1*mean
  - Soil resource cost: distribution from observations

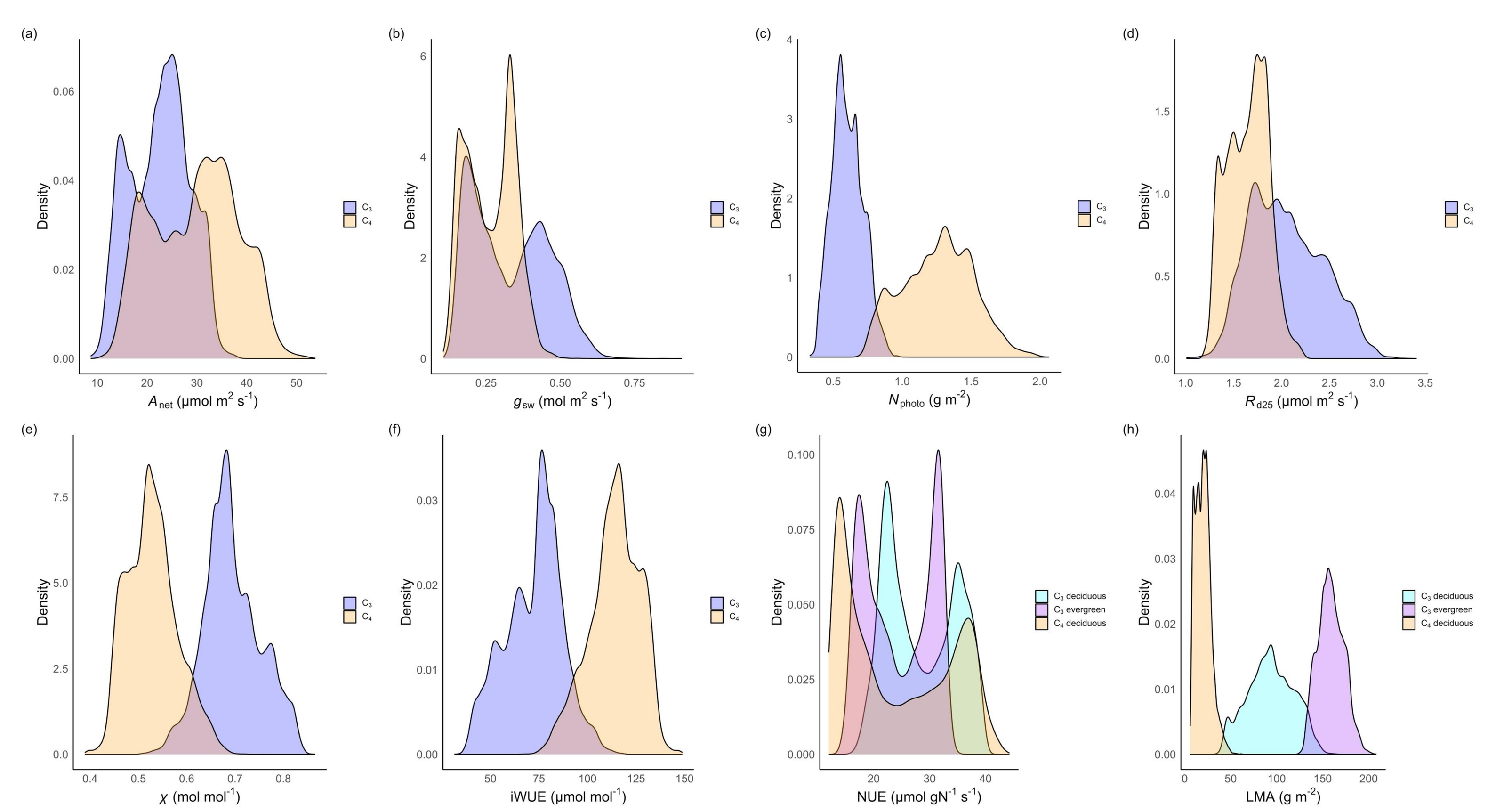



## (3) Global photosynthetic strategies are defined by plant type and climate



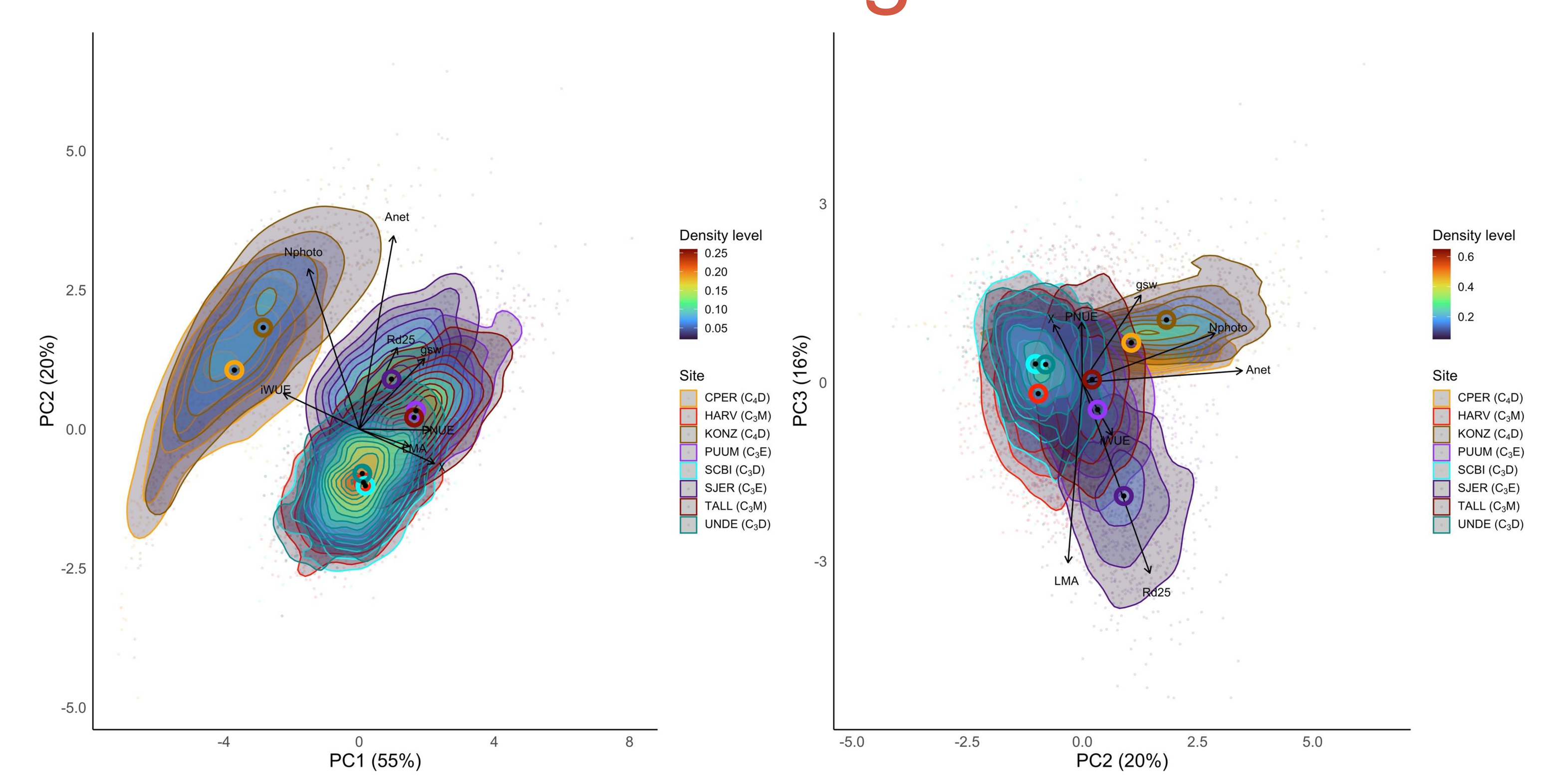
PC1: Water use efficiency (iWUE) x nitrogen use ($N_{photo}$) tradeoff axis
PC2: Nitrogen use efficiency (PNUE) x water use ($g_{sw}$) tradeoff axis, correlated with net carbon assimilation ($A_{net}$)
PC3: Photosynthetic construction and maintenance cost tradeoff axis ($R_{d25}$, LMA)

## (4) Global trait variability is wide



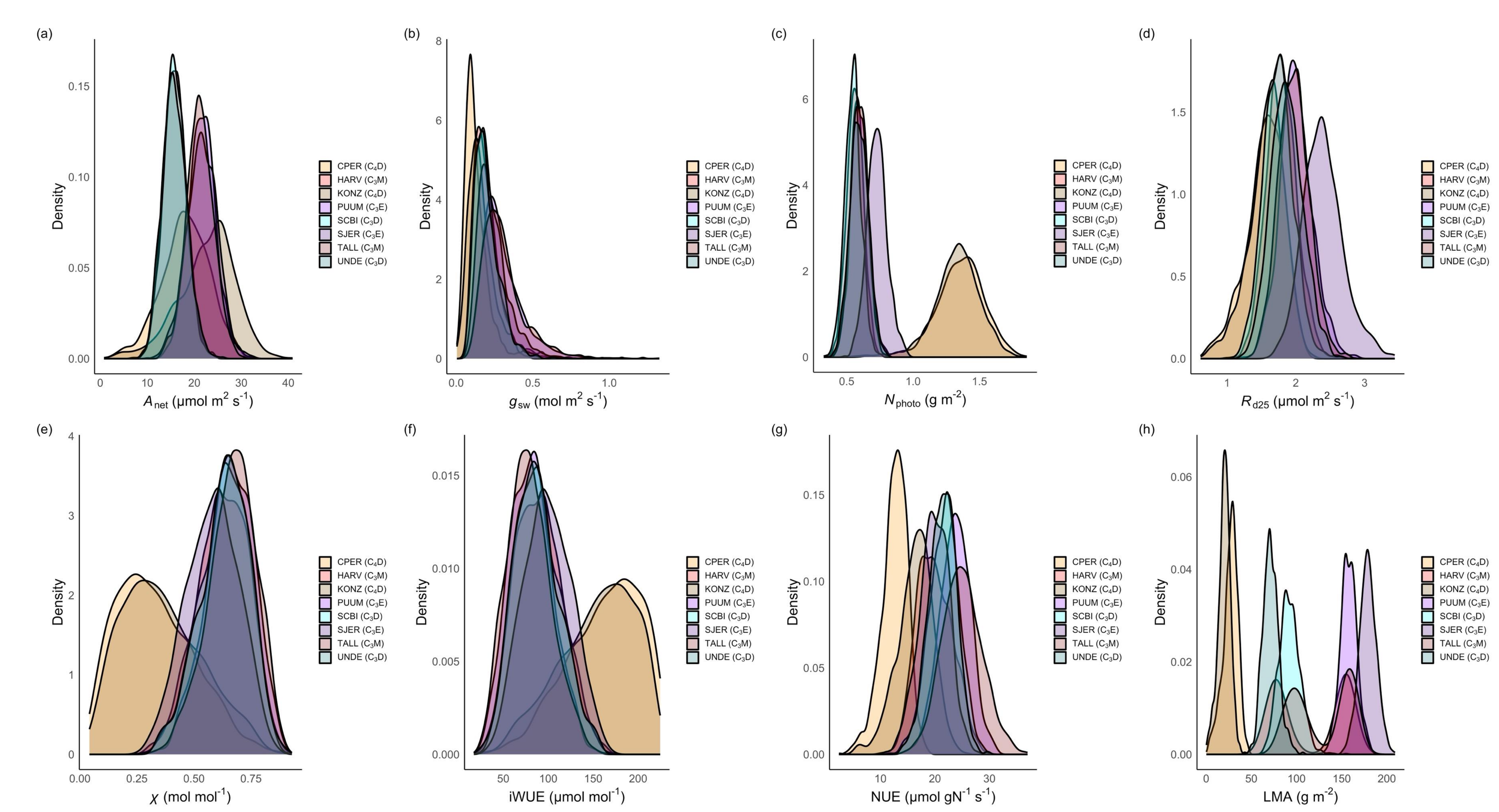
- Climate variability leads to many optimal photosynthetic traits
- Plant type drives separation in some traits, but not others
  - χ due to $C_4$ CCM, but not reflected in $g_{sw}$
  - iWUE reflects high efficiency of $C_4$, relative to $C_3$
  - $N_{photo}$ reflects added enzymatic costs for $C_4$, relative to $C_3$
  - LMA reflects leaf construction costs

## (5) Locally, there is greater within- than across-site variability in strategies



- Principal component separation mimics global analysis
- Within sites, many optimal photosynthetic strategies exist
- Generally, more within- than across-site variability
- Areas where $C_4$ dominate show clear separation in strategies

## (6) Locally, many optimal strategies exist



- Large degree of within-site variability in all optimal traits
- Only strong separation by plant type for $N_{photo}$ and LMA
  - Reflects global analysis
- Mixed sites show intermediate values between evergreen and deciduous sites